

A Correlation-Aware Graph Transformer for Joint Regional Load and Operational Stress Forecasting in Bangladesh

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Abstract—Day-ahead planning in developing-economy national grids requires synchronised visibility over regional electricity demand and system-wide operational pressure, yet load prediction and stress monitoring are often handled separately. On the Bangladesh national power system, daily records from nine administrative divisions jointly report evening-peak load with limitation, reserve, and shedding indicators, making coupled forecasting necessary. We implement the Correlation-Aware Multi-Task Forecasting Framework as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT) that encodes a seven-day lookback with a train-estimated correlation graph and decodes regional demand and a bounded Operational Stress Index (OSI) through shared spatio-temporal encoding and task-specific heads. On a frozen held-out test year, correlation-graph PF-STGT (S2) achieves the strongest joint performance in the internal suite—macro demand MAE 88.65 MW and OSI R^2 0.745—with statistically supported demand improvement over the historical hybrid reference (median Δ MAE -5.43 MW, $p = 5.5 \times 10^{-5}$). Ablation favours correlation adjacency over geographical or hybrid priors and documents a demand-only versus joint-stress trade-off. Post-hoc attribution highlights limitation and calendar coalitions with partial cross-method agreement. Results are scoped to offline chronological evaluation on Bangladesh division-level data rather than demonstrated deployment outcomes.

Index Terms—Multi-task learning, spatio-temporal forecasting, operational stress index, correlation graph, smart grid, Bangladesh

I. INTRODUCTION

National grid operators must anticipate regional load and assess system pressure on a day-ahead horizon to coordinate generation dispatch, reserve planning, and constraint management. In developing economies, supply limitations and load shedding often coincide with rapid demand growth, so planners need forecasts that show both where load will concentrate and how severely operational constraints may bind system-wide. On the Bangladesh national power system, operational data are reported across nine administrative divisions with regional evening-peak demand and limitation dynamics co-visible in each daily record [1]. Effective planning therefore depends on spatially disaggregated forecasts that remain temporally aligned with system-wide stress indicators.

Despite the availability of such data, forecasting practice often treats load prediction and stress monitoring as separate pipelines. Classical and ensemble models [2]–[4] forecast demand without explicit inter-division graph structure; shallow recurrent or graph-convolution architectures [5], [6] may fail to combine long-range temporal context with informative multi-node representations. Graph-based methods from residential or microgrid settings often rely on geographical adjacency that may not capture data-driven co-movement among regional

loads [7]. Operational stress—from reserve shortfall, limitation pressure, and shedding intensity—is seldom forecast jointly with regional demand; shedding studies [8] typically address control rather than day-ahead graded stress indices. Explainability is also rarely integrated with multi-node graph-temporal models that emit coupled load and stress outputs [9], [10].

We address this gap by formulating day-ahead forecasting as a supervised multi-task problem [11] over nine spatial units. Task 1 predicts next-day regional evening-peak demand (MW); Task 2 predicts a graph-level Operational Stress Index (OSI) in $[0, 1]$ summarising system-wide pressure from shedding intensity, generation reserve margin, and operational limitation components. Demand and OSI are operationally related yet statistically non-redundant: demand describes expected load by region, whereas OSI encodes constraint-related pressure not fully determined by demand alone. The central research question is whether a single spatio-temporal framework can learn coupled regional demand and operational stress forecasts from historical Bangladesh grid records under chronological evaluation.

This paper proposes the Correlation-Aware Multi-Task Forecasting Framework, implemented as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT). The framework ingests a seven-day lookback of node-level and national context features with a train-estimated correlation graph, combines graph-aware spatial attention with temporal transformer capacity [12], and decodes regional demand and bounded OSI through task-specific heads with a repaired multi-task loss. Evaluation on chronologically partitioned Bangladesh grid records includes classical, recurrent, and graph-convolution benchmarks, configuration ablations, inferential testing, and post-hoc attribution.

The principal contributions are as follows. First, a correlation-graph PF-STGT for joint day-ahead regional demand and OSI prediction on Bangladesh national grid data. Second, evidence-driven architecture selection favouring correlation-only adjacency over geographical or hybrid priors. Third, a repaired multi-task training protocol with explicit demand–stress loss balancing. Fourth, an empirical evaluation programme with benchmarks, ablations, Wilcoxon testing with Bonferroni correction, and bootstrap confidence intervals. Fifth, integrated explainability with multi-method attribution for both tasks.

Section II presents the methodology; Sections III–?? report experiments and results; Sections IV–V discuss findings and conclude.

II. METHODOLOGY

A. Problem Formulation

Let $\mathcal{V} = \{1, \dots, N\}$ denote $N=9$ administrative divisions. At forecast origin t , the model receives a lookback window of $T=7$ days and predicts targets at horizon $H=1$. Node features $\mathbf{X}^{\text{node}} \in \mathbb{R}^{T \times N \times F_n}$ ($F_n=9$) encode division-level demand, supply, load, lags, rolling means, and regional shares. Global features $\mathbf{X}^{\text{global}} \in \mathbb{R}^{T \times F_g}$ ($F_g=17$) encode calendar indicators, grid aggregates, limitation variables, and national generation

scalars. Task 1 outputs regional demand $\hat{\mathbf{y}}^{\text{demand}} \in \mathbb{R}^N$ (MW); Task 2 outputs graph-level OSI $\hat{y}^{\text{osi}} \in [0, 1]$.

OSI integrates three train-normalised components that capture distinct pressure mechanisms:

$$\begin{aligned} c_1 &= L_{\text{total}}/D_{\text{total}}, \\ c_2 &= 1 - GR/\text{Highest_Gen}, \\ c_3 &= \text{TOL}/\text{Highest_Gen}, \\ \text{OSI} &= \text{mean}(\text{minmax}_{\text{train}}(c_1), \text{minmax}_{\text{train}}(c_2), \\ &\quad \text{minmax}_{\text{train}}(c_3)), \end{aligned} \quad (1)$$

where L_{total} , D_{total} , GR , and TOL denote total shedding, demand, generation reserve, and operational limitation. Component c_1 reflects shedding intensity relative to demand; c_2 captures reserve shortfall; and c_3 encodes infrastructure limitation pressure. Continuous OSI regression preserves graded stress magnitude even when shedding is absent. Same-day OSI is excluded from inputs when forecasting $\text{OSI}(t+1)$ to prevent leakage.

The joint objective combines macro-averaged Huber loss on demand [13] with MSE on OSI:

$$\mathcal{L} = \mathcal{L}_{\text{demand}}^{\text{raw}}/100 + \lambda_2 \mathcal{L}_{\text{osi}}, \quad \lambda_2 = 20. \quad (2)$$

Without demand-loss scaling, megawatt-scale gradients dominate the bounded OSI term and the model collapses to demand-only learning; dividing by 100 MW and setting $\lambda_2=20$ restores comparable optimisation pressure across both heads.

B. Dataset and Graph Construction

We use the Bangladesh Smart Grid dataset (Mendeley Data, Version 5) [1]: 1,850 daily records (Nov. 2019–Dec. 2024) across nine divisions. Chronological partitions yield 1,281 training, 263 validation, and 264 test sliding windows (Table I). All preprocessing parameters—scaling, encoding, graph weights, and OSI bounds—are estimated on training data only.

TABLE I
DATASET PROPERTIES AND S2 TRAINING CONFIGURATION.

Dataset	Value	Training	Value
Divisions (N)	9	Optimiser	Adam, $\text{lr } 5 \times 10^{-4}$
Window (T) / horizon	7 / 1 day	Batch / epochs	32 / ≤ 200
Node / global features	9 / 17	λ_2 (Eq. 2)	20.0
Train / val / test windows	1281 / 263 / 264	Parameters	749,058
Graph edges ($\tau=0.65$)	33 / 36 pairs	Early stop	Patience 15

Inter-regional evening-peak demand on the Bangladesh grid exhibits strong positive co-movement driven by shared national growth and seasonality rather than border adjacency alone. We therefore construct a data-driven spatial prior: pairwise Pearson correlations ρ_{ij} computed on training-partition demand define undirected edges with $\rho_{ij} \geq 0.65$, weighted by ρ_{ij} and row-normalised to form adjacency $\mathbf{A}$. The correlation-only graph retains 33 of 36 possible pairs. Geographical adjacency (21 pairs) and hybrid adjacency (24 pairs) serve as ablation comparators only.

C. PF-STGT Architecture

Fig. 1 shows the end-to-end pipeline; Fig. 2 details configuration S2. Node and global tensors are embedded with regional identity and positional encoding.

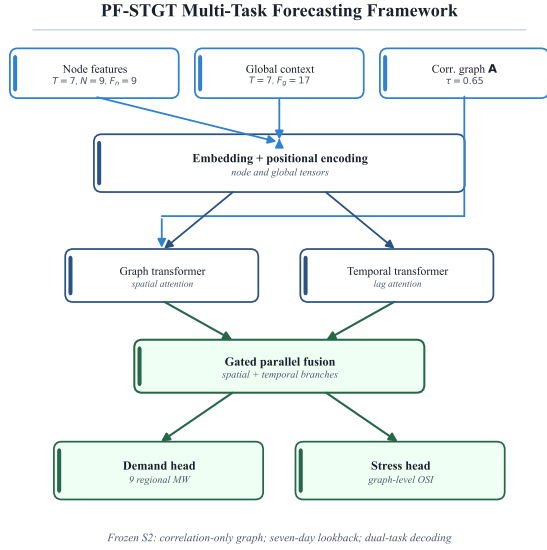


Fig. 1. End-to-end PF-STGT multi-task forecasting framework (frozen S2). Seven-day node and global tensors, a train-estimated correlation graph, parallel spatial-temporal encoding, and dual demand/OSI decoding heads.

A *spatial branch* applies graph-transformer self-attention over nodes at each timestep with additive bias from $\mathbf{A}$:

$$\text{Attn}_{\text{spatial}}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B}_{\text{adj}}\right) \mathbf{V}. \quad (3)$$

A *temporal branch* applies a shared transformer encoder over each node’s seven-day sequence. Learnable gating fuses the branches rather than forcing a fixed spatial-then-temporal ordering, allowing the model to emphasise cross-division synchronisation or within-division persistence depending on context. The final timestep yields shared node embeddings $\mathbf{H}_{\text{shared}}$. A per-node linear head decodes regional demand; a graph-level head combining pooled node states and global context decodes OSI through a bounded activation. Both tasks share encoders and differ only in decoding layers, so limitation and calendar structure learned for stress prediction remains available to the demand head without duplicating feature extraction across separate models.

III. EXPERIMENTAL SETUP

A. Baselines and Metrics

We compare S2 against seven reference models spanning classical machine learning (linear regression, random forest [2], XGBoost [3]), recurrent deep learning (LSTM [5], GRU [14]), graph convolution (T-GCN [6]), and the historical PF-STGT reference B07 (hybrid geographical–correlation graph, W20 protocol). All models share chronological partitions, the frozen feature store, windowing ($T=7$, $H=1$), and leakage policies,

ensuring that performance differences reflect architecture rather than data handling.

Task 1 is evaluated with macro-averaged MAE (primary ranking metric), RMSE, MAPE, and R^2 over nine divisions. Macro MAE assigns equal weight to each division and is preferred over R^2 for planning because absolute megawatt error directly affects dispatch decisions. Task 2 uses MAE and R^2 on OSI. Paired Wilcoxon signed-rank tests [15] on per-window macro demand MAE apply Bonferroni correction [16]; bootstrap 95% confidence intervals [17] and Cohen’s d [18] supplement p -values. Inferential testing targets demand MAE because it is the primary optimisation and selection criterion; OSI performance is reported through point estimates.

B. Training Configuration

S2 trains with Adam [19] ($\eta=5 \times 10^{-4}$, weight decay 10^{-4}), batch size 32, up to 200 epochs with early stopping (patience 15 on composite validation score), and gradient clipping at 1.0. Model capacity: $d_{\text{model}}=128$, four heads, two spatial and two temporal layers (749,058 parameters). Deep baselines use AdamW with up to 60 epochs; classical models use training-partition standardisation only. Seed 42 governs all stochastic operations.

C. Ablation Design

Six PF-STGT configurations (A1–A6) isolate graph topology (none, geographical, hybrid, correlation-only), transformer removal, and single-task demand-only training. A1 equals B07; A6 equals S2. This design permits component-wise attribution of demand and OSI performance to graph prior, temporal encoder, and multi-task formulation.

D. Explainability

Post-hoc attribution [10], [20] on the frozen S2 checkpoint applies grouped GradientSHAP, permutation importance, and spatial attention export without retraining (Experiment 04).

Spatial attention. Fig. 3 visualises mean spatial attention on the train-estimated correlation graph, averaged over 24 case studies. Attribution mass concentrates on Dhaka (340.36), Rajshahi, and Khulna; peripheral divisions receive lower mass. The Spearman correlation between learned attention and frozen adjacency is $\rho=0.422$, indicating moderate alignment between data-driven graph structure and learned spatial coupling.

Coalition attributions. Fig. 4 summarises grouped GradientSHAP magnitudes ($|\phi|$) on coalition registry G1–G11 (validation $n=20$). On the stress task, limitation stack (G8) and calendar/trend (G6) rank highest ($|\phi|=0.0191$ and 0.0190 ; panel (b)). On Dhaka demand, calendar (G6), engineered lags (G4), and national generation (G10) dominate ($|\phi|=162.34$, 101.26 , 91.44 ; panel (a)), indicating task-specific coalition emphasis under shared encoding.

Cross-method check. Dual-path stress comparison agrees in 13 of 24 case studies (52.2%). Demand gradient and permutation rankings disagree ($\rho=-0.564$), whereas stress rankings partially align ($\rho=0.366$). Attributions describe model behaviour on the evaluation sample, not physical causation.

Frozen S2 Architecture — Correlation-Only PF-STGT

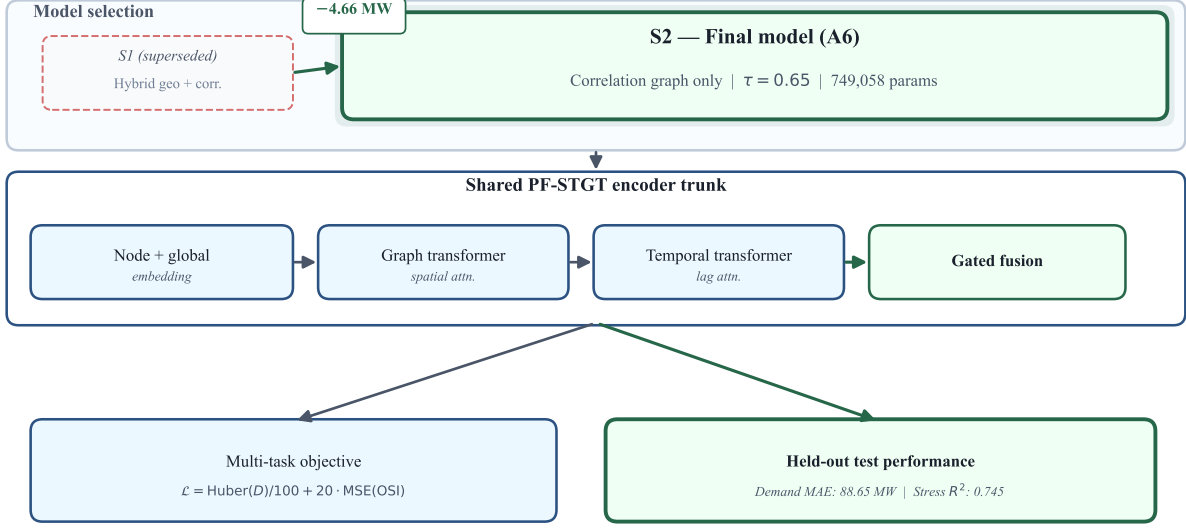


Fig. 2. Frozen S2 (A6) architecture with correlation-only graph ($\tau=0.65$). S2 replaces hybrid-graph S1 and reports held-out demand MAE of 88.65 MW and stress R^2 of 0.745.

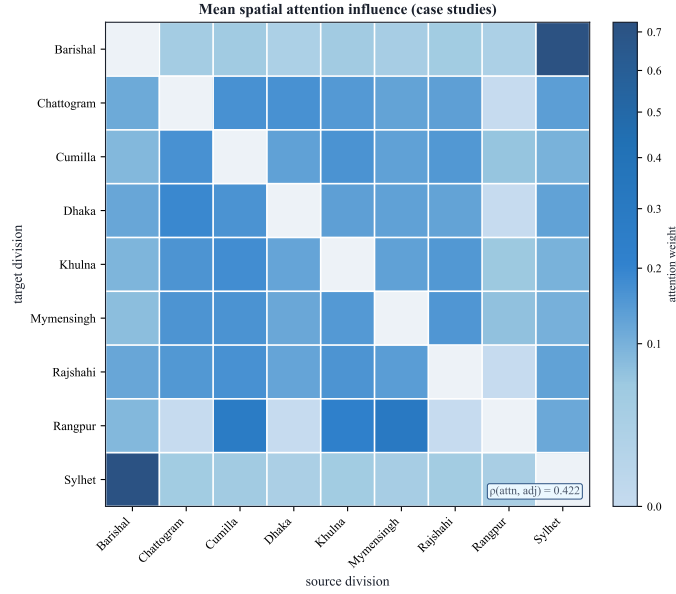


Fig. 3. Mean spatial attention influence on the correlation graph, averaged over 24 case studies. Cell (i, j) gives attention from source division j to target i ; diagonal entries are masked. $\rho(\text{attention}, \text{adjacency})=0.422$.

IV. DISCUSSION

These results demonstrate that one correlation-graph PF-STGT can supply operationally coupled day-ahead load and stress visibility from shared representations on Bangladesh national grid data. The discussion below interprets *why* the observed patterns arise, what they imply for the Bangladesh setting, and how they should inform deployment decisions.

A. Why Correlation Graphs Outperform Geography

Geographical adjacency encodes plausible power-exchange neighbourhoods but assigns equal weight to border links regardless of observed demand coupling. On the Bangladesh national footprint, train-period evening-peak demand correlations are uniformly positive across divisions, reflecting shared national growth trajectories, holiday calendars, and temperature-driven seasonality that propagate system-wide rather than

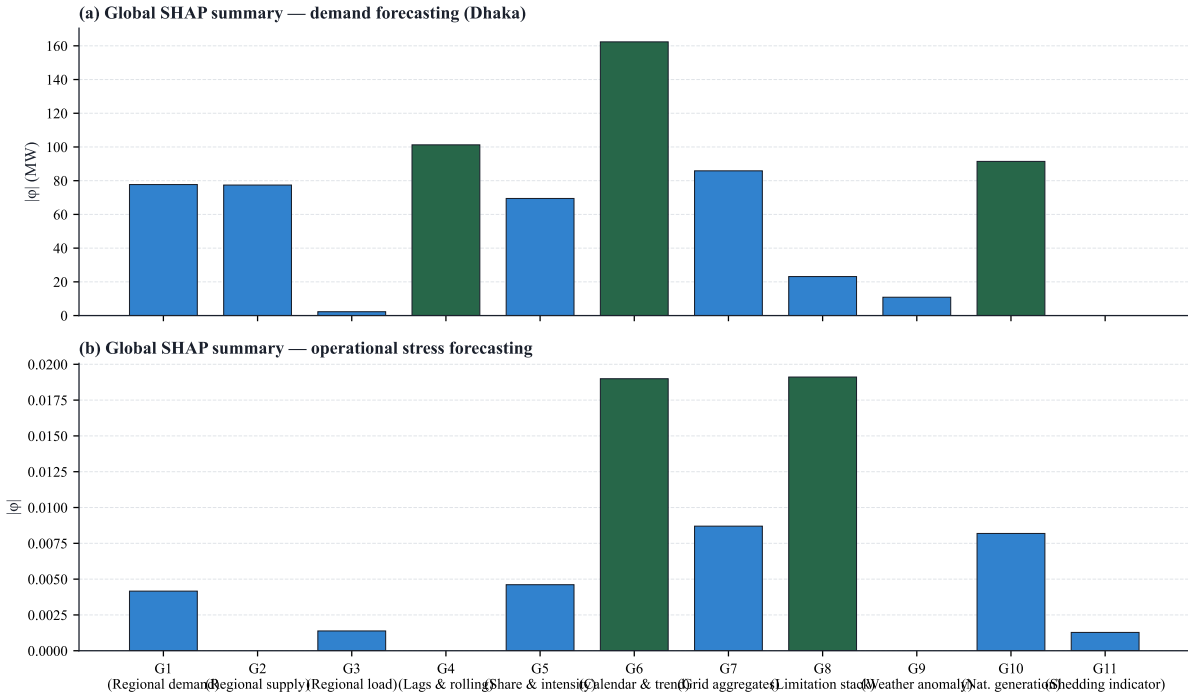


Fig. 4. Grouped GradientSHAP mean absolute attributions ($|\phi|$) on coalition registry G1–G11 (validation $n=20$). (a) Dhaka demand: calendar (G6), lags (G4), and national generation (G10) dominate. (b) Stress: limitation stack (G8) and calendar (G6) rank highest.

along administrative borders. The correlation graph at $\tau=0.65$ retains 33 of 36 pairs because most divisions co-vary strongly; geographical adjacency retains only 21 border-adjacent pairs and therefore under-connects divisions that move together without sharing a border.

A5 (geographical graph) attains marginally lower OSI MAE (0.0340) yet 9.33 MW higher demand error than S2, confirming that stress-optimal and demand-optimal topologies diverge. Geographical conditioning significantly degrades demand ($p=1.48 \times 10^{-4}$) because it restricts spatial message passing to a sparse, equal-weight skeleton that does not reflect the dense co-movement structure the transformer encoder exploits. Correlation weighting, by contrast, supplies graded edge strengths aligned with predictive coupling, which the graph-attention bias in Eq. (3) can refine further at each timestep. This pattern is consistent with developing-economy national grids where inter-regional synchronisation dominates border-local exchange [21], [22].

B. Why Multi-Task Learning Behaves as Observed

Demand and OSI share calendar structure, limitation dynamics, and national generation context but differ in output semantics: regional megawatt levels versus a bounded system-wide stress scalar. Shared spatio-temporal encoding allows the model to learn representations of limitation-relevant structure—visible in attribution rankings for coalition G8 (limitation stack) and G6 (calendar/trend)—that single-task demand models must infer indirectly through global covariates alone. Without repaired loss balancing, megawatt-scale Huber gradients suppress OSI head training; dividing demand loss by 100 MW and

setting $\lambda_2=20$ restores gradient parity so both heads receive comparable optimisation pressure.

The demand-only ceiling (A4, 86.89 MW) is 1.76 MW below S2 because the stress head introduces a regularising signal that slightly offsets pure demand fitting, while joint training enables OSI outputs with $R^2=0.745$. This is not task-interference in the destructive sense but a documented Pareto trade-off: organisations requiring load forecasts alone may prefer A4; centres needing synchronised load and stress visibility should accept the modest demand offset for substantially more informative stress predictions than B07 or classical baselines provide.

C. Bangladesh Grid Implications

Three characteristics of the Bangladesh operational record shape the results. First, nine divisions with co-visible shedding, reserve, and limitation fields in each daily record make composite stress supervision feasible without external telemetry. Second, Dhaka’s dominance ($\sim 35.7\%$ of national demand, highest test-set MAE) means macro-averaged metrics understate capital-region planning sensitivity; error concentration persists across all model families, indicating a data-scale challenge rather than an architecture-specific failure. Third, daily cadence with a seven-day lookback aligns with administrative reporting practice, explaining why shallow recurrent and T-GCN baselines underperform relative to full transformer capacity on the correlation graph: they lack sufficient expressive power to model multi-day seasonal structure within the window while propagating information across densely connected nodes.

D. Practical Deployment Considerations

A single forecasting service can update regional evening-peak trajectories and a national OSI scalar from one seven-day input window on the same daily cycle as the forecast horizon. Operators could review paired forecasts during day-ahead planning, treating Dhaka as a separately reported diagnostic and monitoring limitation-stack and calendar inputs highlighted by attribution. Shadow-mode operation alongside incumbent tools is advisable before cutover, with division-level MAE thresholds and periodic stress-case review substituting for absent prediction intervals. These recommendations are planning diagnostics grounded in offline accuracy—not demonstrated reductions in grid reliability, economic outcome, or deployment success.

E. Limitations and Future Work

Evidence is bounded to a single Bangladesh timeline with daily cadence and frozen internal benchmarks; no external replication or field deployment trial is included. Inferential support applies to demand MAE; OSI gains lack hypothesis tests or prediction intervals. Attribution describes model behaviour on the evaluation sample, not physical causation, with documented cross-method discordance ($\rho = -0.564$ on demand). Uncertainty quantification and S2 residual time-series diagnostics were not executed.

Future work could replicate the chronological protocol on comparable South Asian national systems, evaluate adaptive correlation graphs, extend probabilistic forecasting with quantile or conformal heads, and conduct operator-in-the-loop validation of dual-task explanation dashboards. Such studies would separate architecture-specific gains from country-specific feature geometry without altering the frozen conclusions reported here.

V. CONCLUSION

This study investigated whether a single spatio-temporal framework can jointly forecast next-day regional evening-peak demand and a graph-level Operational Stress Index from Bangladesh national grid records under chronological evaluation. The Correlation-Aware Multi-Task Forecasting Framework—implemented as correlation-graph PF-STGT with parallel spatial and temporal transformer encoding and repaired multi-task loss—delivers the strongest joint credentials in the frozen benchmark suite: macro demand MAE 88.65 MW and OSI R^2 0.745, with statistically supported demand improvement over the historical hybrid reference.

Five outcomes define the contribution. First, one checkpoint supplies coupled day-ahead load and stress forecasts from shared spatio-temporal representations. Second, correlation adjacency outperforms geographical priors on this dataset. Third, a documented demand–stress trade-off exists against demand-only training. Fourth, regional error geography is predictable, with meaningful Dhaka gains despite persistent concentration at the national load centre. Fifth, multi-method attribution provides coalition-level transparency with explicit discordance reporting.

The work extends spatio-temporal forecasting to coupled stress indices on a previously unreported Bangladesh division-level case [1], demonstrating that graph priors and task weighting require empirical tuning rather than domain intuition. Coupled day-ahead load-and-stress forecasting is achievable as a single-model service when graph structure, multi-task calibration, and division-level monitoring are co-designed with the evaluation evidence—appropriately qualified as offline evaluation rather than operational deployment.

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